

Logistic Regression

Fitting a model

Dependent variable is chd (yes=coronary heart disease developed, no=coronary heart disease was not developed).

Odds ratios are below, cigs variable is significant.

If you are using Posit R Cloud instead of RStudio Desktop, please use the command below to download the data otherwise use `library(faraway);data(wcgs,package="faraway")`; codes to download the data.

```
tmp <- tempfile(fileext = ".rda")
download.file(
  "https://raw.githubusercontent.com/cran/faraway/master/data/wcgs.rda",
  destfile = tmp,
  mode = "wb"
)
load(tmp)
head(wcgs)
```

##	age	height	weight	sdp	dbp	chol	behave	cigs	dibep	chd	typechd	timechd	arcus
## 1	49	73	150	110	76	225	A2	25	A	no	none	1664	absent
## 2	42	70	160	154	84	177	A2	20	A	no	none	3071	present
## 3	42	69	160	110	78	181	B3	0	B	no	none	3071	absent
## 4	41	68	152	124	78	132	B4	20	B	no	none	3064	absent
## 5	59	70	150	144	86	255	B3	20	B	yes	infdeath	1885	present
## 6	44	72	204	150	90	182	B4	0	B	no	none	3102	absent

```
#library(faraway)
#data(wcgs, package="faraway")
lmod <- glm(chd ~ height + cigs, family = binomial, wcgs)
summary(lmod)
```

```
##
## Call:
## glm(formula = chd ~ height + cigs, family = binomial, data = wcgs)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -4.50161    1.84186  -2.444   0.0145 *
## height      0.02521    0.02633   0.957   0.3383
## cigs        0.02313    0.00404   5.724 1.04e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1781.2  on 3153  degrees of freedom
## Residual deviance: 1749.0  on 3151  degrees of freedom
## AIC: 1755
##
```

```
## Number of Fisher Scoring iterations: 5
```

```
beta <- coef(lmod)
```

The Deviance is the deviance for the current model while the Null Deviance is the deviance for a model with no predictors and just an intercept term.

Null deviance: 1781.2 - Residual deviance: 1749.0 = 32

This test statistic is asymptotically distributed assuming that the smaller model is correct and the distributional assumptions hold.

For example, we can compare the fitted model to the null model (which has no predictors) by considering the difference between the residual and null deviances. For the heart disease example, this difference is 32.2 on two degrees of freedom (one for each predictor). Hence, the p-value for the test of the hypothesis that at least one of the predictors is related to the response is:

Ho: Variables are not important

Ha: Variable are important

```
1-pchisq(1781.2 - 1749, 2)
```

```
## [1] 1.01826e-07
```

Since this value is so small, we are confident that there is some relationship between the predictors and the response. Reject Ho. Accept Ha.

We can say that the odds of heart disease increase by 2.6% with each additional inch in height and by 2.3% with each additional cigarette smoked per day. Note that $\exp(x) \approx 1+x$ for small values of x. We observe that so the $\beta_2 = 2.3\%$ increase in odds due to smoking a cigarette a day could have been quickly estimated from the original model output without further computation. This approximation is useful for quick intuitions. It is more natural to compute the effect of a pack a day (20 cigarettes):

```
exp(beta[3])
```

```
##      cigs
```

```
## 1.023397
```

```
exp(beta[3]*20)
```

```
##      cigs
```

```
## 1.588115
```

So we have 59% increase in the odds of heart disease due to smoking.

The model is significant based on the anova table above. Can we drop height?

Ho: Height can be dropped

Ha: Height can not be dropped.

```
anova(lmod, test="Chi") # it is logistic regression, you need to specify that it is Chi Square test
```

```
## Analysis of Deviance Table
```

```
##
```

```
## Model: binomial, link: logit
```

```
##
```

```
## Response: chd
```

```
##
```

```
## Terms added sequentially (first to last)
```

```
##
```

```
##
```

```
##          Df Deviance Resid. Df Resid. Dev Pr(>Chi)
```

```
## NULL          3153      1781.2
## height  1      1.125      3152      1780.1      0.2888
## cigs    1     31.070      3151      1749.0 2.49e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
lmodc <- glm(chd ~ cigs, family = binomial, wchs)
anova(lmodc, lmod, test="Chi")
```

```
## Analysis of Deviance Table
##
## Model 1: chd ~ cigs
## Model 2: chd ~ height + cigs
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         3152         1750
## 2         3151         1749  1  0.92025   0.3374
```

```
#alternatively
drop1(lmod, test="Chi") #this GLM Test for each variable
```

```
## Single term deletions
##
## Model:
## chd ~ height + cigs
##           Df Deviance    AIC    LRT Pr(>Chi)
## <none>          1749.0 1755.0
## height  1      1750.0 1754.0  0.9202   0.3374
## cigs    1      1780.1 1784.1 31.0695 2.49e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
lmode <- glm(chd ~ height, family = binomial, wchs)
anova(lmode, lmod, test="Chi")
```

```
## Analysis of Deviance Table
##
## Model 1: chd ~ height
## Model 2: chd ~ height + cigs
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         3152         1780.1
## 2         3151         1749.0  1    31.07 2.49e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Accept Ho. Yes, height can be dropped from the model. 95% confidence intervals for the coefficients

```
confint(lmod)
```

```
## Waiting for profiling to be done...
```

```
##           2.5 %      97.5 %
## (Intercept) -8.13475465 -0.91297018
## height      -0.02619902  0.07702835
## cigs         0.01514949  0.03100534
```

```
exp(confint(lmod))
```

```
## Waiting for profiling to be done...
```

```
##              2.5 %    97.5 %
## (Intercept) 0.000293171 0.4013304
## height      0.974141198 1.0800727
## cigs        1.015264825 1.0314910
```

Prediction:

```
test<-data.frame(cbind(height=72,cigs=0))
predict(lmod,test,type="response",se.fit=TRUE)
```

```
## $fit
##      1
## 0.06376553
##
## $se.fit
##      1
## 0.006434148
##
## $residual.scale
## [1] 1
```

Goodness of Fit Test:

Ho: The model is a good fit Ha: The model is not a good fit.

```
library(ResourceSelection)
```

```
## ResourceSelection 0.3-6    2023-06-27
```

```
hoslem.test(lmod$y,fitted(lmod),g=5)
```

```
##
## Hosmer and Lemeshow goodness of fit (GOF) test
##
## data:  lmod$y, fitted(lmod)
## X-squared = 2.8341, df = 3, p-value = 0.4179
```

Accept Ho. Fit is good.

Model Selection

```
#lets add BMI to the data and then perform variable selection
wcgs$bmi <- with(wcgs, 703*wcgs$weight/(wcgs$height^2))
lmod <- glm(chd ~ age + height + weight +bmi + sdp + dbp + chol + dibep + cigs +arcus, family=binomial,
lmodr <- step(lmod, trace=0)
summary(lmodr)
```

```
##
## Call:
## glm(formula = chd ~ age + height + bmi + sdp + chol + dibep +
##      cigs + arcus, family = binomial, data = wcgs)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -15.299983   2.294134  -6.669 2.57e-11 ***
## age           0.061590   0.012397   4.968 6.76e-07 ***
## height       0.050161   0.027824   1.803  0.0714 .
```

```
## bmi            0.060385    0.026599    2.270    0.0232 *
## sdp            0.017728    0.004155    4.267 1.98e-05 ***
## chol          0.010709    0.001529    7.006 2.45e-12 ***
## dibepB       -0.657616    0.145898   -4.507 6.56e-06 ***
## cigs          0.021041    0.004262    4.936 7.96e-07 ***
## arcuspresent  0.210998    0.143718    1.468    0.1421
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1769.2 on 3139 degrees of freedom
## Residual deviance: 1569.3 on 3131 degrees of freedom
## (14 observations deleted due to missingness)
## AIC: 1587.3
##
## Number of Fisher Scoring iterations: 6
```

arcuspresent and height are not significant. Lets drop them one at a time, starting with the variable with the highest p value

```
drop1(lmodr,test="Chi")
```

```
## Single term deletions
##
## Model:
## chd ~ age + height + bmi + sdp + chol + dibep + cigs + arcus
##      Df Deviance   AIC    LRT Pr(>Chi)
## <none>      1569.3 1587.3
## age      1   1593.9 1609.9 24.598 7.062e-07 ***
## height   1   1572.6 1588.6  3.277  0.07028 .
## bmi      1   1574.4 1590.4  5.098  0.02396 *
## sdp      1   1586.6 1602.6 17.285 3.217e-05 ***
## chol     1   1620.0 1636.0 50.658 1.099e-12 ***
## dibep    1   1590.7 1606.7 21.320 3.886e-06 ***
## cigs     1   1592.8 1608.8 23.453 1.280e-06 ***
## arcus    1   1571.5 1587.5  2.130  0.14441
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
lmodr1<-glm(formula = chd ~ age + height + bmi + sdp + chol + dibep +
  cigs , family = binomial, data = wcgs)
summary(lmodr1)
```

```
##
## Call:
## glm(formula = chd ~ age + height + bmi + sdp + chol + dibep +
##      cigs, family = binomial, data = wcgs)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -15.688095    2.283610  -6.870 6.43e-12 ***
## age          0.064808    0.012099   5.357 8.48e-08 ***
## height       0.055351    0.027681   2.000  0.0455 *
## bmi          0.055911    0.026522   2.108  0.0350 *
## sdp          0.017592    0.004119   4.271 1.95e-05 ***
```

```
## chol          0.011062    0.001520    7.279 3.36e-13 ***
## dibepB        -0.653174    0.145167   -4.499 6.81e-06 ***
## cigs          0.020957    0.004244    4.938 7.90e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1779.2  on 3141  degrees of freedom
## Residual deviance: 1580.6  on 3134  degrees of freedom
## (12 observations deleted due to missingness)
## AIC: 1596.6
##
## Number of Fisher Scoring iterations: 6
drop1(lmodr1,test="Chi")
```

```
## Single term deletions
##
## Model:
## chd ~ age + height + bmi + sdp + chol + dibep + cigs
##      Df Deviance    AIC    LRT Pr(>Chi)
## <none>      1580.5 1596.5
## age      1  1609.1 1623.1 28.553 9.119e-08 ***
## height   1  1584.6 1598.6  4.036  0.04454 *
## bmi      1  1585.0 1599.0  4.398  0.03598 *
## sdp      1  1597.9 1611.9 17.308 3.179e-05 ***
## chol     1  1635.3 1649.3 54.712 1.396e-13 ***
## dibep    1  1601.8 1615.8 21.227 4.079e-06 ***
## cigs     1  1604.0 1618.0 23.451 1.281e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

All variables are significant.

Confusion Matrix

Lets use 0.5 as a cutoff for the model to identify if a patient has developed coronary heart disease.

```
library(caret)

## Loading required package: ggplot2
## Loading required package: lattice

#convert defaults from "Yes" and "No" to 1's and 0's
true_labels<- ifelse(wcgs$chd=="yes", 1, 0)
predicted_probabilities <- predict(lmodr1, wcgs, type="response")

#using 0.5 as a cutoff to predict the class
predictions <- ifelse(predicted_probabilities > 0.5, 1, 0)
confusion_matrix <- confusionMatrix(as.factor(predictions),as.factor(true_labels),mode="prec_recall", p
confusion_matrix

## Confusion Matrix and Statistics
##
##              Reference
```

```
## Prediction      0      1
##              0 2882  255
##              1   3   2
##
##              Accuracy : 0.9179
##              95% CI : (0.9077, 0.9273)
##      No Information Rate : 0.9182
##      P-Value [Acc > NIR] : 0.5425
##
##              Kappa : 0.0122
##
##      McNemar's Test P-Value : <2e-16
##
##              Precision : 0.4000000
##              Recall : 0.0077821
##              F1 : 0.0152672
##              Prevalence : 0.0817950
##      Detection Rate : 0.0006365
##      Detection Prevalence : 0.0015913
##      Balanced Accuracy : 0.5033711
##
##      'Positive' Class : 1
##
```

#using 0.4 as a cutoff to predict the class

```
predictions <- ifelse(predicted_probabilities > 0.4, 1, 0)
confusion_matrix <- confusionMatrix(as.factor(predictions),as.factor(true_labels),mode="prec_recall", p
confusion_matrix
```

```
## Confusion Matrix and Statistics
##
##              Reference
## Prediction      0      1
##              0 2864  247
##              1   21  10
##
##              Accuracy : 0.9147
##              95% CI : (0.9044, 0.9242)
##      No Information Rate : 0.9182
##      P-Value [Acc > NIR] : 0.7742
##
##              Kappa : 0.0528
##
##      McNemar's Test P-Value : <2e-16
##
##              Precision : 0.322581
##              Recall : 0.038911
##              F1 : 0.069444
##              Prevalence : 0.081795
##      Detection Rate : 0.003183
##      Detection Prevalence : 0.009866
##      Balanced Accuracy : 0.515816
##
##      'Positive' Class : 1
##
```

```
#using 0.1 as a cutoff to predict the class
predictions <- ifelse(predicted_probabilities > 0.1, 1, 0)
confusion_matrix <- confusionMatrix(as.factor(predictions),as.factor(true_labels),mode="prec_recall", p
confusion_matrix
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##           0 2192  100
##           1   693  157
##
##           Accuracy : 0.7476
##           95% CI : (0.732, 0.7627)
##   No Information Rate : 0.9182
##   P-Value [Acc > NIR] : 1
##
##           Kappa : 0.1807
##
##   McNemar's Test P-Value : <2e-16
##
##           Precision : 0.18471
##           Recall : 0.61089
##           F1 : 0.28365
##           Prevalence : 0.08180
##   Detection Rate : 0.04997
##   Detection Prevalence : 0.27053
##   Balanced Accuracy : 0.68534
##
##   'Positive' Class : 1
##
```

```
#using 0.08 as a cutoff to predict the class
predictions <- ifelse(predicted_probabilities > 0.08, 1, 0)
confusion_matrix <- confusionMatrix(as.factor(predictions),as.factor(true_labels),mode="prec_recall", p
confusion_matrix
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##           0 1967   69
##           1   918  188
##
##           Accuracy : 0.6859
##           95% CI : (0.6693, 0.7021)
##   No Information Rate : 0.9182
##   P-Value [Acc > NIR] : 1
##
##           Kappa : 0.165
##
##   McNemar's Test P-Value : <2e-16
##
##           Precision : 0.16998
##           Recall : 0.73152
```

```
##          F1 : 0.27586
##          Prevalence : 0.08180
##          Detection Rate : 0.05983
##          Detection Prevalence : 0.35201
##          Balanced Accuracy : 0.70666
##
##          'Positive' Class : 1
##
```

```
#ROC Curve
```

```
prob<-predicted_probabilities
library(pROC)
```

```
## Type 'citation("pROC")' for a citation.
```

```
##
```

```
## Attaching package: 'pROC'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

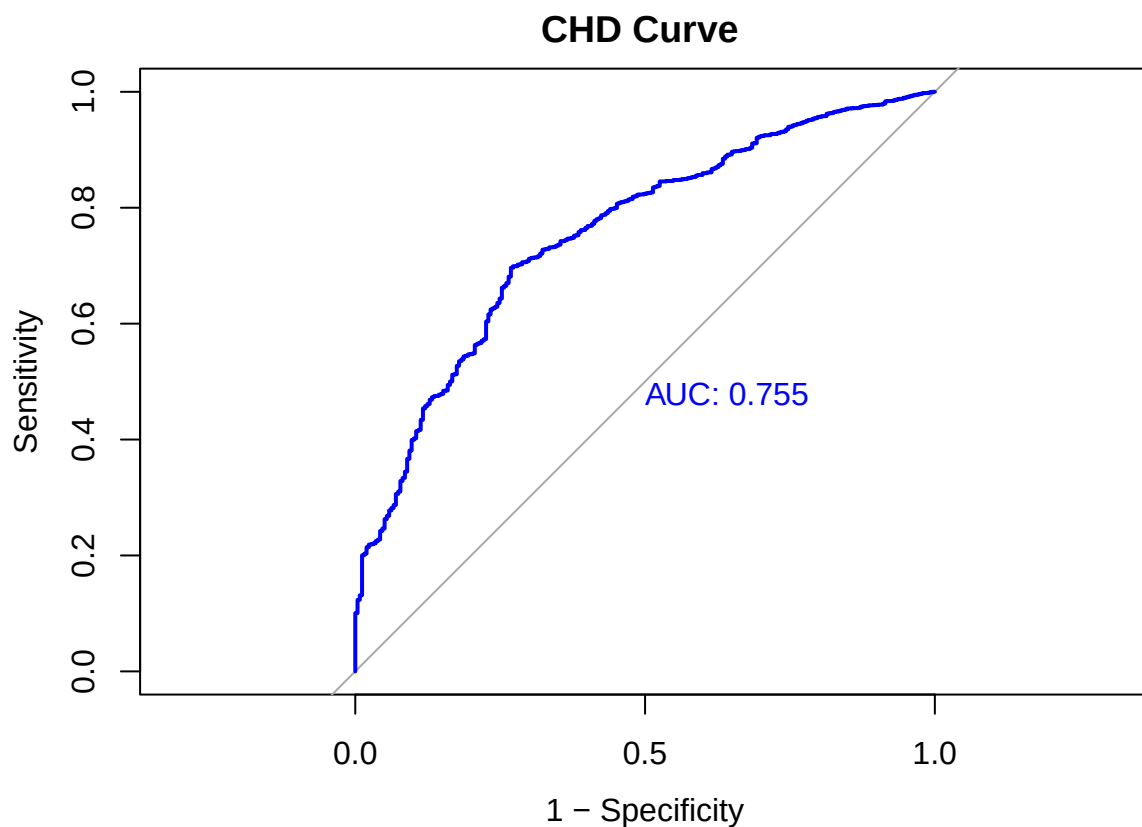
```
##      cov, smooth, var
```

```
par(mfrow=c(1,1))
```

```
credit_roc <- roc(wcgs$chd,prob,levels=c("yes", "no"))
```

```
## Setting direction: controls > cases
```

```
plot.roc(credit_roc,print.auc=TRUE,main = "CHD Curve ", col = "blue", lwd = 2, legacy.axes = TRUE)
```



```
#r code to find the optimal cutoff based on max F1 score
```

```

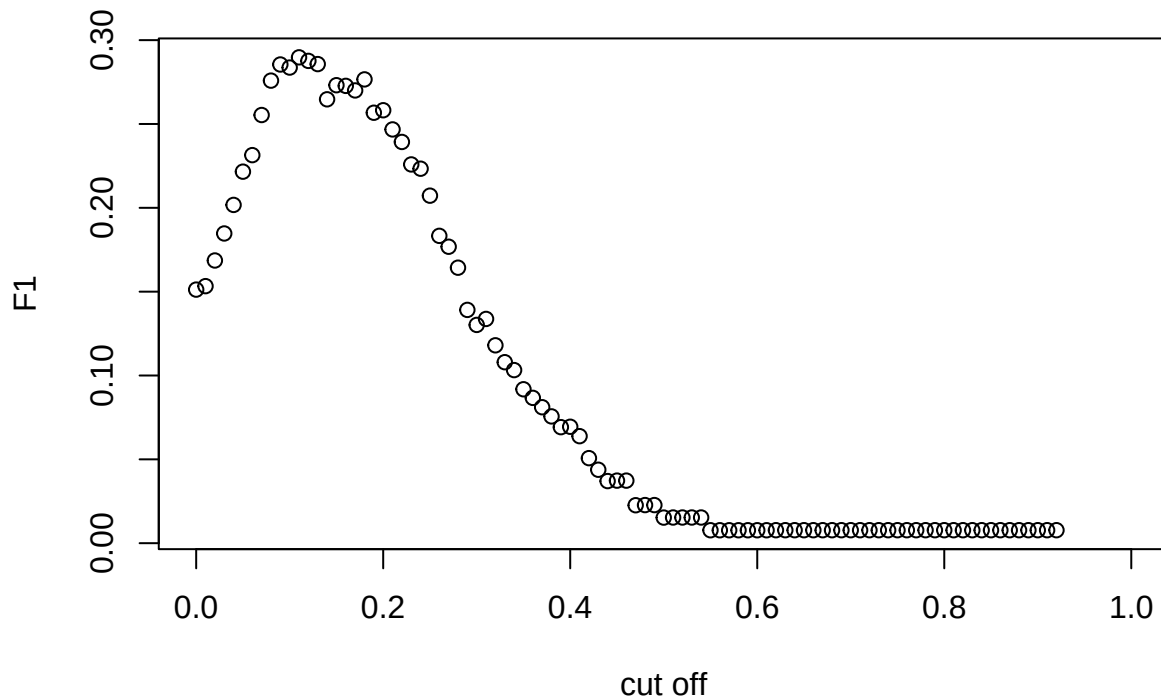
cutoff.prg<-function(pred,act){
  # pred<-predicted_probabilities
  # act<-true_labels
  p<-seq(0,1,0.01)
  n<-length(p)
  out<-matrix(0,nrow=n,ncol=12)
  for(i in 1:n){
    predictions <- ifelse(pred >p[i], 1, 0)
    confusion_matrix <- confusionMatrix(as.factor(predictions),as.factor(act),mode="prec_recall", positive = "1")
    out[i,]<-cbind(p=p[i],t(confusion_matrix[[4]]))
  }
  dimnames(out)[[2]]<-c("p","Sensitivity","Specificity","Pos Pred Value","Neg Pred Value","Precision","Recall")
  out
}

test<-cutoff.prg(predicted_probabilities,true_labels)

## Warning in confusionMatrix.default(as.factor(predictions), as.factor(act), :
## Levels are not in the same order for reference and data. Refactoring data to
## match.
## Warning in confusionMatrix.default(as.factor(predictions), as.factor(act), :
## Levels are not in the same order for reference and data. Refactoring data to
## match.
## Warning in confusionMatrix.default(as.factor(predictions), as.factor(act), :
## Levels are not in the same order for reference and data. Refactoring data to
## match.
## Warning in confusionMatrix.default(as.factor(predictions), as.factor(act), :
## Levels are not in the same order for reference and data. Refactoring data to
## match.
## Warning in confusionMatrix.default(as.factor(predictions), as.factor(act), :
## Levels are not in the same order for reference and data. Refactoring data to
## match.
## Warning in confusionMatrix.default(as.factor(predictions), as.factor(act), :
## Levels are not in the same order for reference and data. Refactoring data to
## match.
## Warning in confusionMatrix.default(as.factor(predictions), as.factor(act), :
## Levels are not in the same order for reference and data. Refactoring data to
## match.
## Warning in confusionMatrix.default(as.factor(predictions), as.factor(act), :
## Levels are not in the same order for reference and data. Refactoring data to
## match.
## Warning in confusionMatrix.default(as.factor(predictions), as.factor(act), :
## Levels are not in the same order for reference and data. Refactoring data to
## match.
## Warning in confusionMatrix.default(as.factor(predictions), as.factor(act), :
## Levels are not in the same order for reference and data. Refactoring data to
## match.

plot(test[,1],test[,8],xlab="cut off",ylab="F1")

```



```
test[which.max(test[,8]),]
```

##	p	Sensitivity	Specificity
##	0.11000000	0.56031128	0.79445407
##	Pos Pred Value	Neg Pred Value	Precision
##	0.19538670	0.95301455	0.19538670
##	Recall	F1	Prevalence
##	0.56031128	0.28973843	0.08179504
##	Detection Rate	Detection Prevalence	Balanced Accuracy
##	0.04583068	0.23456397	0.67738268

#finding optimal point based on equality of sensitivity and specificity

```
plot(test[,1],
      test[,8],
      type = "l",
      col = 2,
      xlab = "cut off",
      ylab = "rates")

# Add line graphs of other two data set
lines(test[,1],
      test[,2],
      type = "l",
      col = 3)
lines(test[,1],
      test[,3],
      type = "l",
      col = 4)

# Add legend in top right corner
legend("topright",
      c("F1", "Sensitivity", "Specificity"),
      lty = 1,
```

```
col = 2:4)
```

